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SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE

Abstract

A semiconductor device includes: a first row line extending in a first direction; a first column line extending in a second direction that intersects the first direction; a first memory cell connected between the first row line and the first column line; a second column line located above the first column line and extending in the second direction; and a first resistance pattern located between the first column line and the second column line.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2025-0001418 filed on Jan. 6, 2025, which is incorporated herein by reference in its entirety.
BACKGROUND
1. Technical Field
[0002] Embodiments of the present disclosure relate to an electronic device, and more particularly, to a semiconductor device and a method of manufacturing the semiconductor device.
2. Related Art
[0003] The degree of integration of a semiconductor device is mainly determined by an area occupied by a unit memory cell. Recently, as the improvement in the degree of integration of a semiconductor device for forming memory cells in a single layer on a substrate reaches a limit, a three-dimensional semiconductor device for stacking memory cells on a substrate has been proposed. Furthermore, in order to improve the operational reliability of such a semiconductor device, various structures and manufacturing methods have been developed.
SUMMARY
[0004] In an embodiment, a semiconductor device may include: a first row line extending in a first direction; a first column line extending in a second direction that intersects the first direction; a first memory cell connected between the first row line and the first column line; a second column line located above the first column line and extending in the second direction; and a first resistance pattern located between the first column line and the second column line.
[0005] In an embodiment, a method of manufacturing a semiconductor device may include: forming a cell line that extends in a first direction and includes a variable resistance layer; forming a first conductive layer above the cell line; forming a hard mask pattern above the first conductive layer, the hard mask pattern extending in a second direction that intersects the first direction; forming a first column line by etching the first conductive layer using the hard mask pattern, the first column line extending in the second direction; forming a first memory cell by etching the cell line; forming liner patterns on sidewalls of the first memory cell, sidewalls of the first column line, and sidewalls of the hard mask pattern; forming an opening between the liner patterns by removing the hard mask pattern; forming a resistance pattern in the opening; and forming a second column line in the opening.
[0006] In an embodiment, a method of manufacturing a semiconductor device may include: forming cell lines that extends in a first direction; forming a first conductive layer on the cell lines; forming hard mask patterns on the first conductive layer, the hard mask patterns extending in a second direction that intersects the first direction; forming first column lines by etching the first conductive layer using the hard mask patterns, the first column lines extending in the second direction;
[0007] forming first memory cells by etching the cell lines using the hard mask patterns, the first memory cells being arranged in the first direction and the second direction; forming gap-fill insulating layers between the first column lines adjacent to each other in the first direction; surface-treating the gap-fill insulating layers; forming openings between the gap-fill insulating layers by removing the hard mask patterns; and forming second column lines in the openings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIGS. **1**A, **1**B, and **1**C are diagrams illustrating the structure of a semiconductor device in

accordance with an embodiment.
[0009] FIGS. **2**A and **2**B are diagrams illustrating the structure of a semiconductor device in accordance with an embodiment.
[0010] FIGS. **3**A, **4**A, **5**A, **6**A, and **7**A, FIGS. **3**B, **4**B, **5**B, **6**B, and **7**B, and FIGS. **3**C, **4**C, **5**C, **6**C, and **7**C are diagrams for describing a method of manufacturing a semiconductor device in accordance with an embodiment.
[0011] FIGS. **8**A, **8**B, **8**C, **8**D, and **8**E are diagrams for describing a method of manufacturing a semiconductor device in accordance with an embodiment.
[0012] FIGS. **9**A, **9**B, and **9**C are diagrams for describing a method of manufacturing a semiconductor device in accordance with an embodiment.
DETAILED DESCRIPTION
[0013] Various embodiments are directed to a semiconductor device having a stable structure and improved characteristics and a method of manufacturing the semiconductor device.
[0014] By stacking memory cells in three dimensions, it is possible to improve the degree of integration of a semiconductor device. It is also possible to provide a semiconductor device having a stable structure and improved reliability.
[0015] Hereafter, embodiments in accordance with the technical spirit of the present disclosure will be described with reference to the accompanying drawings. Throughout the specification and claims, a list of items prefaced by a phrase such as "at least one of" or "one or more of" or "one or both of" indicates an inclusive list. For example, a list of "at least one of A or B" and a list of "one or both of A and B" each indicate A, or B, or AB (i.e., A and B). Moreover, a first element "on" a second element indicates that the first element can be "directly on" the second element, or that at least one intervening element can be interposed between the first and second elements.
[0016] FIGS. **1**A to **1**C are diagrams illustrating the structure of a semiconductor device in accordance with an embodiment. FIG. **1**A is a plan view, FIG. **1**B is a cross-sectional view taken along line A-A′ of FIG. **1**A, and FIG. **1**C is a cross-sectional view taken along line B-B′ of FIG. **1**A.
[0017] Referring to FIGS. **1**A to **1**C, the semiconductor device may include a first row line RL**1**, a first column line CL**1**, a first memory cell MC**1**, a second column line CL**2**, and a first resistance pattern R**1**. The semiconductor device may further include at least one of a second row line RL**2**, a second memory cell MC**2**, a second resistance pattern R**2**, a third resistance pattern R**3**, a fourth resistance pattern R**4**, a fifth resistance pattern R**5**, a first liner pattern L**1**, a second liner pattern L**2**, third liner patterns L**3**A and L**3**B, a fourth liner pattern L**4**, a first gap-fill insulating layer GF**1**, a second gap-fill insulating layer GF**2**, a third gap-fill insulating layer GF**3**, or a fourth gap-fill insulating layer GF**4**.
[0018] The first row line RL**1** and the second row line RL**2** may extend in a first direction I, and the first column line CL**1** and the second column line CL**2** may extend in a second direction II intersecting the first direction I. The first row line RL**1**, the first column line CL**1**, the second column line CL**2**, and the second row line RL**2** may be sequentially stacked along a third direction III. The third direction III may be a direction perpendicular to a plane defined by the first direction I and the second direction II. The first and second row lines RL**1** and RL**2** may be word lines, and the first and second column lines CL**1** and CL**2** may be bit lines. Alternatively, the first and second row lines RL**1** and RL**2** may be bit lines, and the first and second column lines CL**1** and CL**2** may be word lines.
[0019] The first memory cell MC**1** may be connected between the first row line RL**1** and the first column line CL**1**, and the second memory cell MC**2** may be connected between the second row line RL**2** and the second column line CL**2**. The first memory cell MC**1** may be a resistive memory cell including a variable resistance layer **13**. Each of the first memory cells MC**1** may include a select element, or a memory element, or both. As an example, each of the first memory cells MC**1** may include a first electrode **11**, a second electrode **12**, and a variable resistance layer **13**, and the

variable resistance layer **13** may be located between the first electrode **11** and the second electrode **12**.

[0020] The variable resistance layer **13** may have characteristics that it reversibly transitions between different resistance states depending on a voltage or a current applied to the first memory cell MC**1**. As an example, when the variable resistance layer **13** has a low resistance state, data ‘1’ may be stored, and when the variable resistance layer **13** has a high resistance state, data ‘0’ may be stored.

[0021] As an example, the variable resistance layer **13** may include a resistive material. An electrical path is generated or disappears in the variable resistance layer **13**, such that data may be stored. As an example, the variable resistance layer **13** may include transition metal oxide or include metal oxide such as a perovskite-based material.

[0022] As an example, the variable resistance layer **13** may have a magnetic tunnel junction (MTJ) structure including a magnetization pinned layer, a tunnel barrier layer, and a magnetization free layer. The data may be stored according to a change in magnetization direction of the magnetization free layer with respect to a magnetization direction of the magnetization pinned layer. As an example, the magnetization pinned layer and the magnetization free layer may each include a magnetic material, and the tunnel barrier layer may include metal oxide.

[0023] As an example, the variable resistance layer **13** may include a phase change material or include a chalcogenide-based material. The variable resistance layer **13** may change its phase according to a program operation. As an example, the variable resistance layer **13** may have a low resistance crystalline state through a set operation. As an example, the variable resistance layer **13** may have a high resistance amorphous state through a reset operation. Accordingly, the data may be stored in the memory cell using a resistance difference depending on a phase of the variable resistance layer **13**.

[0024] As an example, the variable resistance layer **13** may include a variable resistance material whose resistance changes without a phase change or include a chalcogenide-based material. The variable resistance layer **13** may maintain its phase after the program operation. As an example, the variable resistance layer **13** may have an amorphous state, and may maintain the amorphous state without changing to a crystalline state after the program operation. A threshold voltage of the memory cell may be changed depending on a program voltage applied to the memory cell, and the memory cell may be programmed to at least two states. As an example, the memory cell may be programmed to a set state or a reset state using program voltages having different polarities. Accordingly, the data may be stored in the memory cell using a difference in the threshold voltage of the memory cell.

[0025] The first electrode **11** may be electrically connected to the first row line RL**1**. The second electrode **12** may be electrically connected to the first column line CL**1**. The first electrode **11** and the second electrode **12** may each include polysilicon, tungsten (W), tungsten nitride (WN.sub.x), tungsten silicide (WSi.sub.x), titanium (Ti), titanium nitride (TiN.sub.x), titanium silicon nitride (TiSiN), titanium aluminum nitride (TiAlN), tantalum (Ta), tantalum nitride (TaN), tantalum silicon nitride (TaSiN), tantalum aluminum nitride (TaAlN), carbon (C), silicon carbide (SIC), silicon carbonitride (SiCN), aluminum (Al), copper (Cu), zinc (Zn), nickel (Ni), cobalt (Co), lead (Pb), platinum (Pt), molybdenum (Mo), ruthenium (Ru), or the like, or include one or more combinations thereof. The first row line RL**1** and the first column line CL**1** may each include metal such as tungsten (W) or molybdenum (Mo).

[0026] The second memory cells MC**2** may each have a similar structure to the first memory cells MC**1**. Each of the second memory cells MC**2** may include a first electrode **14**, a second electrode **15**, and a variable resistance layer **16**, and the variable resistance layer **16** may be located between the first electrode **14** and the second electrode **15**.

[0027] The first liner patterns L**1** may be located on sidewalls of the first memory cells MC**1** adjacent to each other in the second direction II, and may extend along sidewalls of the first row

lines RL**1**. The first gap-fill insulating layers GF**1** may be located between the first memory cells MC**1** adjacent to each other in the second direction II, and may extend between the first row lines RL**1**. For example, each of the first gap-fill insulating layers GF**1** may be located between a pair of the first memory cells MC**1** adjacent to each other in the second direction II, and may extend between a pair of the first row lines RL**1** adjacent to each other in the second direction II.
[0028] The second liner patterns L**2** may be located on sidewalls of the first memory cells MC**1** adjacent to each other in the first direction I, and may extend along sidewalls of the first column lines CL**1**. The second liner pattern L**2** may extend along a sidewall of the second column line CL**2**, and may cover a portion of the sidewall of the second column line CL**2**. Specifically, the second liner pattern L**2** may cover a sidewall of a corresponding (e.g., abutting) one of the first memory cells MC**1**, a sidewall of a corresponding (e.g., abutting) one of the first column lines CL**1**, and a portion of a sidewall of a corresponding (e.g., adjacent to) one of the second column lines CL**2**. For example, the second liner pattern L**2** may cover a pair of sidewalls of an adjacent pair of the first memory cell MC**1** in the first direction I, a pair of sidewalls of an adjacent pair of the first column lines CL**1** in the first direction I, and portions of sidewalls of an adjacent pair of the second column lines CL**2** in the first direction I. The second gap-fill insulating layers GF**2** may be located between the first memory cells MC**1** adjacent to each other in the first direction I, and may extend between the first column lines CL**1**. For example, each of the second gap-fill insulating layers GF**2** may be located between a pair of the first memory cells MC**1** adjacent to each other in the first direction I, and may extend between a pair of the first column lines CL**1** adjacent to each other in the first direction I.
[0029] The third liner patterns L**3**A may be located on sidewalls of the second memory cells MC**2** adjacent to each other in the first direction I. The third liner patterns L**3**B may be located on the third liner patterns L**3**A, and may extend along sidewalls of the second column lines CL**2**. The third gap-fill insulating layers GF**3** may be located between the second memory cells MC**2** adjacent to each other in the first direction I, and may extend between the second column lines CL**2**.
[0030] The fourth liner patterns L**4** may be located on sidewalls of the second memory cells MC**2** adjacent to each other in the second direction II, and may extend along sidewalls of the second row lines RL**2**. The fourth gap-fill insulating layers GF**4** may be located between the second memory cells MC**2** adjacent to each other in the second direction II, and may extend between the second row lines RL**2**.
[0031] The first to fourth liner patterns L**1** to L**4** may each include nitride. The first to fourth gap-fill insulating layers GF**1** to GF**4** may each include oxide.
[0032] The first resistance pattern R**1** may be located between the first column line CL**1** and the second column line CL**2**. The second column line CL**2** may be located above the first column line CL**1**, and the first resistance pattern R**1** may extend in the second direction II. For example, the first resistance pattern R**1** may extend along an upper surface of the first column line CL**1** and a lower surface of the second column line CL**2**. The second column line CL**2** may be electrically connected to the first column line CL**1** through the first resistance pattern R**1**.
[0033] The first resistance pattern R**1** is used to reduce damage to the memory cells MC**1** and MC**2** due to a spike current, and may be a high resistivity pattern. The first resistance pattern R**1** may include a material having a higher resistivity than the first and second column lines CL**1** and CL**2**. As an example, the first and second column lines CL**1** and CL**2** may each include tungsten (W), molybdenum (Mo), or the like, and the first resistance pattern R**1** may include at least one of tungsten nitride (WN), tungsten silicon nitride (WSIN), molybdenum nitride (MoN), molybdenum silicon nitride (MoSiN), titanium nitride (TiN), or titanium chloride (TiCl.sub.4).
[0034] The first resistance pattern R**1** may be located between the first column line CL**1** and the second column line CL**2** and between the second column line CL**2** and the second liner patterns L**2**. The first resistance pattern R**1** may partially cover the sidewall of the second column line CL**2**. The second column line CL**2** may include a first portion P**1** and a second portion P**2**. The first resistance

pattern R1 may cover a bottom surface and a sidewall of the first portion P1. The first resistance pattern R1 may not cover the second portion P2. The second portion P2 may be covered by the third liner pattern L3B. For example, the first resistance pattern R1 may cover a pair of opposing sidewalls of the first portion P1 that are arranged in the first direction I and a bottom surface of the first portion P1.

[0035] The first portion P1 and the second portion P2 may have different widths. The first portion P1 may have a first width W1, the second portion P2 may have a second width W2, and the second width W2 may be greater than the first width W1. The first width W1 and the second width W2 may be different from each other by a thickness of the first resistance pattern R1. As the first width W1 decreases by the thickness of the first resistance pattern R1, an area associated with contact resistance between the first column line CL1 and the second column line CL2 may decrease compared to that without including the first resistance pattern R1 between the first and second column lines CL1 and CL2. In addition, the resistivity of the first resistance pattern R1 may be greater than that of each of the first and second column lines CL1 and CL2. As a result, contact resistance between the first column line CL1 and the second column line CL2 may increase compared to that without including the first resistance pattern R1 between the first and second column lines CL1 and CL2. Accordingly, the damage to the memory cells MC1 and MC2 due to the spike current may be further reduced.

[0036] The second resistance pattern R2 may be located between the first memory cells MC1 and the first column line CL1, and may extend in the second direction II along a lower surface of the first column line CL1. The third resistance pattern R3 may be located between the second memory cells MC2 and the second column line CL2, and may extend in the second direction II along an upper surface of the second column line CL2. The fourth resistance pattern R4 may be located between the first row line RL1 and the first memory cells MC1, and may extend in the first direction I along an upper surface of the first row line RL1. The fifth resistance pattern R5 may be located between the second row line RL2 and the second memory cells MC2, and may extend in the first direction I along a lower surface of the second row line RL2.

[0037] The second to fifth resistance patterns R2 to R5 are used to reduce damage to the memory cells MC1 and MC2 due to a spike current, and may be relatively high resistivity patterns. The second to fifth resistance patterns R2 to R5 may each include a material having a higher resistivity than the first and second column lines CL1 and CL2 and/or the first and second row lines RL1 and RL2. As an example, the second to fifth resistance patterns R2 to R5 may each include at least one of tungsten nitride (WN), tungsten silicon nitride (WSiN), molybdenum nitride (MoN), molybdenum silicon nitride (MoSiN), titanium nitride (TiN), or titanium chloride (TiCl.sub.4). According to the structure described above, a first memory cell array including the first memory cells MC1 and a second memory cell array including the second memory cells MC2 may share the first and second column lines CL1 and CL2 with each other. The semiconductor device may reduce the damage to the memory cells MC1 and MC2 due to the spike current by including at least one of the first to fifth resistance patterns R1 to R5.

[0038] FIGS. 2A and 2B are diagrams illustrating the structure of a semiconductor device in accordance with an embodiment. FIG. 2A is a cross-sectional view in the first direction I, and FIG. 2B is a cross-sectional view in the second direction II. Hereinafter, the content overlapping with the previously described content may be omitted for the interest of brevity.

[0039] Referring to FIGS. 2A and 2B, the semiconductor device may include a first row line RL1, a first column line CL1, a first memory cell MC1, a second column line CL2, and a first resistance pattern R1. The semiconductor device may further include at least one of a second row line RL2, a second memory cell MC2, a second resistance pattern R2, a third resistance pattern R3, a fourth resistance pattern R4, a fifth resistance pattern R5, a first liner pattern L1, a second liner pattern L2, a third liner pattern L3, a fourth liner pattern L4, a first gap-fill pattern GF1, a second gap-fill insulating layer GF2, a third gap-fill insulating layer GF3, or a fourth gap-fill insulating layer GF4.

[0040] The first memory cell MC**1** may be connected between the first row line RL**1** and the first column line CL**1**, and the second memory cell MC**2** may be connected between the second row line RL**2** and the second column line CL**2**. The first memory cell MC**1** may include a first electrode **21**, a second electrode **22**, and a variable resistance layer **23**. The second memory cell MC**2** may include a first electrode **24**, a second electrode **25**, and a variable resistance layer **26**.

[0041] The first liner patterns L**1** and the first gap-fill insulating layers GF**1** may be located between the first memory cells MC**1** adjacent to each other in the second direction II. The first liner patterns L**1** and the first gap-fill insulating layers GF**1** may extend along sidewalls of the first row lines RL**1**. The fourth liner patterns L**4** and the fourth gap-fill insulating layers GF**4** may be located between the second memory cells MC**2** adjacent to each other in the second direction II. The fourth liner patterns L**4** and the fourth gap-fill insulating layers GF**4** may extend along sidewalls of the second row lines RL**2**.

[0042] The second liner patterns L**2** may be located on sidewalls of first memory cells MC**1** adjacent to each other in the first direction I. The second liner patterns L**2** may extend along sidewalls of the first column lines CL**1** and sidewalls of the second column lines CL**2**. In some embodiments, each of the second liner patterns L**2** may cover a sidewall (e.g., substantially an entire sidewall) of a corresponding (e.g., abutting) one of the first memory cells MC, a sidewall (e.g., substantially an entire sidewall) of a corresponding (e.g., abutting) one of the first column lines CL**1**, and a sidewall (e.g., substantially an entire sidewall) of a corresponding (e.g., adjacent to) one of the second column lines CL**2**. The second gap-fill insulating layers GF**2** may be located between the first memory cells MC**1** adjacent to each other in the first direction I. The second gap-fill insulating layers GF**2** may extend between the first column lines CL**1** and between the second column lines CL**2**. The third liner patterns L**3** and the third gap-fill insulating layers GF**3** may be located between the second memory cells MC**2** adjacent to each other in the first direction I.

[0043] The first resistance pattern R**1** may be located between the first column line CL**1** and the second column line CL**2** and between the second column line CL**2** and the second liner patterns L**2**. The first resistance pattern R**1** may cover a lower surface and the sidewall of the second column line CL**2**. The first resistance pattern R**1** may extend in the second direction II along an upper surface of the first column line CL**1** and a lower surface of the second column line CL**2**, and may extend along the sidewall of the second column line CL**2**.

[0044] The first resistance pattern R**1** may entirely cover the sidewall of the second column line CL**2**. For example, the first resistance pattern R**1** may substantially entirely cover a pair of opposing sidewalls of the second column line CL**2** that are arranged in the first direction I and a bottom surface of the second column line CL**2**. An upper surface of the second column line CL**2** and an upper surface of the first resistance pattern R**1** may be located at substantially the same level. In other words, an upper surface of the second column line CL**2** and an upper surface of the first resistance pattern R**1** may be substantially coplanar with each other. The upper surface of the second column line CL**2** and an upper surface of the second gap-fill insulating layer GF**2** may be located at substantially the same level. For example, a level difference between an upper surface of the second column line CL**2** and an upper surface of the first resistance pattern R**1** may be not greater than 5%, or 3%, or 1% of a height of the second column line CL**2** in the third direction III.

[0045] The first column line CL**1** and the second column line CL**2** may have different widths. The first column line CL**1** may have a first width WA, the second column line CL**2** may have a second width WB, and the second width WB may be smaller than the first width WA. The first width WA and the second width WB may be different from each other by a thickness of the first resistance pattern R**1**. As the second width WB decreases by the thickness of the first resistance pattern R**1**, an area associated with contact resistance between the first column line CL**1** and the second column line CL**2** may decrease compared to that without including the first resistance pattern R**1** between the first and second column lines CL**1** and CL**2**. In addition, the resistivity of the first resistance pattern R**1** may be greater than that of each of the first and second column lines CL**1** and CL**2**. As a result, resistance between the first column line CL**1** and the second column line CL**2** may increase compared to that without including the first resistance pattern R**1** between the first and second column lines CL**1** and CL**2**. Accordingly, the damage to the memory cells MC**1** and MC**2** due to the spike current may be further reduced.

[0046] The second resistance pattern R**2** may be located between the first memory cells MC**1** and the first column line CL**1**, the third resistance pattern R**3** may be located between the second memory cells MC**2** and the second column line CL**2**, the fourth resistance pattern R**4** may be located between the first row line RL**1** and the first memory cells MC**1**, and the fifth resistance pattern R**5** may be located between the second row line RL**2** and the second memory cells MC**2**.

[0047] According to the structure described above, a first memory cell array including the first memory cells MC**1** and a second memory cell array including the second memory cells MC**2** may share the first and second column lines CL**1** and CL**2** with each other. The semiconductor device may reduce the damage to the memory cells MC**1** and MC**2** due to the spike current by including at least one of the first to fifth resistance patterns R**1** to R**5**.

[0048] FIGS. **3**A, **4**A, **5**A, **6**A, and **7**A, FIGS. **3**B, **4**B, **5**B, **6**B, and **7**B, and FIGS. **3**C, **4**C, **5**C, **6**C, and **7**C are diagrams for describing a method of manufacturing a semiconductor device in accordance with an embodiment. FIGS. **3**A, **4**A, **5**A, **6**A, and **7**A are plan views, FIGS. **3**B, **4**B, **5**B, **6**B, and **7**B are cross-sectional views taken along lines C-C′ of FIGS. **3**A, **4**A, **5**A, **6**A, and **7**A, respectively, and FIGS. **3**C, **4**C, **5**C, **6**C, and **7**C are cross-sectional views taken along lines D-D′ of FIGS. **3**A, **4**A, **5**A, **6**A, and **7**A, respectively. Hereinafter, the content overlapping with the previously described content may be omitted for the interest of brevity.

[0049] Referring to FIGS. **3**A to **3**C, first row lines **30** and first cell lines CE**1** extending in the first direction I are formed. The first cell line CE**1** may include a first electrode **32**, a first variable resistance layer **33**, and a second electrode **34**. A resistance pattern **31** may be located between the first row line **30** and the first cell line CE**1**. The resistance pattern **31** may extend in the first direction I between the first row line **30** and the first cell line CE**1**.

[0050] Subsequently, first liner patterns **35** may be formed on sidewalls of the first cell lines CE**1** and the first row lines **30**. Subsequently, first gap-fill insulating layers **36** may be formed between the first cell lines CE**1** adjacent to each other in the second direction II.

[0051] Referring to FIGS. **4**A to **4**C, a resistance layer, a first conductive layer, and a hard mask layer may be formed above the first cell lines CE**1** and the first gap-fill insulating layers **36**. For example, the first conductive layer may be formed on the resistance layer and above the first cell lines CE**1**, and the hard mask layer may be formed on the first conductive layer. Subsequently, hard mask patterns **39** extending in the second direction II may be formed on the first conductive layer by etching the hard mask layer. Subsequently, first column lines **38** extending in the second direction II may be formed by etching the first conductive layer using the hard mask patterns **39** as etching barriers. Subsequently, a resistance pattern **37** may be formed by etching the resistance layer, and first memory cells MC**1** may be formed by etching the first cell lines CE**1**.

[0052] Subsequently, second liner patterns **40** may be formed on sidewalls of the first memory cells MC**1**, the resistance patterns **37**, the first column lines **38**, and the hard mask patterns **39**. Specifically, a pair of adjacent second liner patterns **40** in the first direction I may be formed on a pair of sidewalls of a corresponding one (e.g., abutting) of the first memory cells MC**1**, a pair of sidewalls of a corresponding one (e.g., abutting) of the first column lines **38**, and a pair of sidewalls of a corresponding one (e.g., abutting) of the hard mask patterns **39**. The second liner patterns **40** may each include a different material from the hard mask pattern **39**. As an example, the hard mask pattern **39** may include a material having a high etching selectivity with respect to the second liner patterns **40**. The second liner patterns **40** may each include nitride, and the hard mask pattern **39** may include polysilicon or oxide.

[0053] Subsequently, second gap-fill insulating layers **41** may be formed between the first memory cells MC**1** and between the first column lines **38** that are adjacent to each other in the first direction

I. The second gap-fill insulating layers **41** may extend in the second direction II between the hard mask patterns **39** adjacent to each other in the first direction I.

[0054] Referring to FIGS. **5**A to **5**C, openings OP may be formed between the second liner patterns **40** by removing the hard mask patterns **39**. For example, each of the openings OP may be formed by removing a corresponding one of the hard mask patterns **39** between an adjacent pair of the second liner patterns **40** in the first direction I. The hard mask patterns **39** may be selectively etched using a difference in etching selectivity between the hard mask patterns **39** and the second liner patterns **40** and a difference in etching selectivity between the hard mask patterns **39** and the second gap-fill insulating layers **41**. As an example, the hard mask patterns **39** may be wet-etched using a dip-out process. The opening OP may extend in the second direction II, and the first column lines **38** may be exposed through the opening OP.

[0055] Referring to FIGS. **6**A to **6**C, a resistance layer **42** may be formed along a surface of the first column line **38** and surfaces of the second liner patterns **40** exposed through the opening OP. The resistance layer **42** may be formed along a profile of the opening OP.

[0056] Subsequently, a second conductive layer **43** is formed. The second conductive layer **43** may be formed to fill the opening OP. Subsequently, a resistance layer **44** may be formed on the second conductive layer **43**.

[0057] Subsequently, a cell stack may be formed on the resistance layer **44**, and hard mask patterns **48** may be formed on the cell stack. The hard mask patterns **48** may be located to correspond to the first column lines **38**, and may extend in the second direction II. Subsequently, second cell lines CE**2** may be formed by etching the cell stack using the hard mask patterns **48** as etching barriers. Each of the second cell lines CE**2** may extend in the second direction II, and may include a first electrode **45**, a variable resistance layer **46**, and a second electrode **47**.

[0058] Subsequently, third liner patterns **49**A may be formed on sidewalls of the hard mask patterns **48** and sidewalls of the second cell lines CE**2**.

[0059] Referring to FIGS. **7**A to **7**C, the resistance layer **44**, the second conductive layer **43**, and the resistance layer **42** are etched using the hard mask patterns **48** as etching barriers. Through this, a resistance pattern **44**A, a second column line **43**A, and a resistance pattern **42**A that extend in the second direction II may be formed. The second column line **43**A may include a first portion located inside the opening OP and a second portion located outside the opening OP and having a greater width than the first portion.

[0060] The resistance pattern **42**A may be located between the first column line **38** and the second column line **43**A, and may cover the first portion of the second column line **43**A. The resistance pattern **42**A may include a material having a higher resistivity than the first and second column lines **38** and **43**A. As an example, the first and second column lines **38** and **43**A may each include tungsten (W), molybdenum (Mo), or the like, and the resistance pattern **42**A may include at least one of tungsten nitride (WN), tungsten silicon nitride (WSiN), molybdenum nitride (MoN), molybdenum silicon nitride (MoSiN), titanium nitride (TiN), or titanium chloride (TiCl.sub.4).

[0061] Subsequently, a liner layer and a gap-fill insulating layer may be formed, and a planarization process may be performed so that the second cell lines CE**2** are exposed. Through this, third liner patterns **49**B and third gap-fill insulating layers **51** may be formed.

[0062] Subsequently, a resistance layer and a third conductive layer may be formed above the second cell lines CE**2**. Subsequently, second row lines **53** extending in the first direction I may be formed by etching the third conductive layer, and resistance patterns **52** may be formed by etching the resistance layer. Subsequently, second memory cells MC**2** arranged in the first direction I and the second direction II may be formed by etching the second cell lines CE**2**.

[0063] Subsequently, fourth liner patterns **54** may be formed on sidewalls of the second row lines **53** and sidewalls of the second memory cells MC**2**, and fourth gap-fill insulating layers **55** may be formed.

[0064] According to the method described above, the second column line **43**A may be formed using the opening OP formed by removing the hard mask pattern **39**. Accordingly, it is possible to reduce damage to the second liner patterns **40** in a process of forming the second memory cells MC**2** and the second column lines **43**A. Specifically, the hard mask patterns **39** are selectively removed compared to the second liner patterns **40** and the second gap-fill insulating layers **41**. As a result, the first column lines **38** may be exposed without damaging the second liner patterns **40**. The second liner patterns **40** may not be damaged by the source gas used to etch the second conductive layer **43**. Because the resistance pattern **42**A is formed in the opening OP, a width of the first portion of the second column line **43**A may be reduced. Accordingly, contact resistance between the first column line **38** and the second column line **43**A may be increased, and damage to the first and second memory cells MC**1** and MC**2** due to a spike current may be reduced.

[0065] FIGS. **8**A to **8**E are diagrams for describing a method of manufacturing a semiconductor device in accordance with an embodiment. Hereinafter, the content overlapping with the previously described content may be omitted for the interest of brevity.

[0066] Referring to FIG. **8**A, first row lines **60** and resistance patterns **61** extending in the first direction I may be formed. First column lines **68**, hard mask patterns **69**, and resistance patterns **67** extending in the second direction II may be formed. First memory cells MC**1** arranged in the first direction I and the second direction II may be formed. The first memory cell MC**1** may include a first electrode **62**, a variable resistance layer **63**, and a second electrode **64**. Second liner patterns **70** and second gap-fill insulating layers **71** may be formed. The second liner pattern **70** may be formed on a sidewall of the first memory cell MC**1**, a sidewall of the first column line **68**, and a sidewall of the hard mask pattern **69**.

[0067] Regions where second column lines are to be formed in a subsequent process may be secured through the hard mask patterns **69**. Accordingly, a height H of the hard mask patterns **69** may be adjusted according to a target height of the second column line.

[0068] Referring to FIG. **8**B, openings OP may be formed between the second liner patterns **70** by removing the hard mask pattern **69**. The openings OP may be located between the second gap-fill insulating layers **71** adjacent to each other in the first direction I. The first column lines **68** may be exposed by the openings OP.

[0069] For reference, the second liner patterns **70** may be partially etched in a process of etching the hard mask pattern **69**. In such a case, upper surfaces of the second liner patterns **70** may be located lower than upper surfaces of the second gap-fill insulating layers **71**, and sidewalls of the second gap-fill insulating layers **71** may be exposed by the openings OP.

[0070] Referring to FIG. **8**C, a resistance pattern **72** and a second column line **73** may be formed in the opening OP. A resistance layer may be formed along surfaces of the first column lines **68**, the second liner patterns **70**, and the second gap-fill insulating layers **71** exposed through the openings OP, and a conductive layer filling the openings OP may be formed on the resistance layer. Subsequently, the resistance patterns **72** and the second column lines **73** may be formed by planarizing the conductive layer and the resistance layer so that the surfaces of the second gap-fill insulating layers **71** are exposed. A planarization process may be performed by a chemical mechanical polishing (CMP) method. An upper surface of the second column line **73**, the upper surface of the second liner pattern **70**, and the upper surface of the second gap-fill insulating layer **71** may be located at substantially the same level.

[0071] In some embodiments (e.g., the embodiment in FIG. **8**C), the upper surfaces of the second liner patterns **70** may be located lower than the upper surfaces of the second gap-fill insulating layers **71**. In such a case, the resistance pattern **72** may cover the upper surfaces of the second liner patterns **70**. In addition, upper surfaces of the resistance pattern **72**, the upper surface of the second column line **73**, and the upper surfaces of the second gap-fill insulating layers **71** may be located at substantially the same level.

[0072] Referring to FIG. **8**D, resistance patterns **74** and second cell lines CE**2** extending in the second direction II may be formed. Each of the second cell lines CE**2** may include a first electrode

75, a variable resistance layer **76**, and a second electrode **77**. Subsequently, third liner patterns **79** and third gap-fill insulating layers **81** may be formed.

[0073] Referring to FIG. **8**E, a resistance layer and a third conductive layer may be formed above the second cell lines CE**2** and the third gap-fill insulating layers **81**. Subsequently, second row lines **83** and resistance patterns **82** extending in the first direction I may be formed by etching the third conductive layer and the resistance layer. Subsequently, second memory cells MC**2** arranged in the first direction I and the second direction II may be formed by etching the second cell lines CE**2**.

[0074] According to the method described above, the resistance pattern **72** may be formed between the first column line **68** and the second column line **73**, and the second column line **73** having a smaller width than the first column line **68** may be formed. Accordingly, contact resistance between the first column line **68** and the second column line **73** may be increased, and a spike current may be reduced.

[0075] FIGS. **9**A to **9**C are diagrams for describing a method of manufacturing a semiconductor device in accordance with an embodiment. Hereinafter, the content overlapping with the previously described content may be omitted for the interest of brevity.

[0076] Referring to FIG. **9**A, first row lines **90** and resistance patterns **91** extending in the first direction I may be formed. First column lines **98**, hard mask patterns **99**, and resistance patterns **97** extending in the second direction II may be formed. First memory cells MC**1** arranged in the first direction I and the second direction II may be formed. The first memory cell MC**1** may include a first electrode **92**, a variable resistance layer **93**, and a second electrode **94**.

[0077] Subsequently, a second liner layer **100** and second gap-fill insulating layers **101** may be formed. The second liner layer **100** may be formed along surfaces (e.g., sidewalls) of the first memory cells MC**1**, surfaces (e.g., sidewalls) of the first column lines **98**, and surfaces (e.g., sidewalls and upper surfaces) of the hard mask patterns **99**. Each of the second gap-fill insulating layers **101** may be formed between a pair of the first column lines **98** adjacent to each other in the first direction I. In addition, the second liner layer **100** may be formed along sidewalls and upper surfaces of the hard mask patterns **99**.

[0078] Subsequently, a surface treatment process may be performed. The surface treatment process may be performed on the second liner layer **100** and the second gap-fill insulating layers **101**. Portions of the second liner layer **100** formed on the upper surfaces of the hard mask patterns **99** may be surface-treated. Upper surfaces of the second gap-fill insulating layers **101** may be surface-treated. As an example, the surface treatment process may be performed using an inert gas such as argon (Ar) or helium (He). The portion T that is surface-treated may have a higher density than a portion that is not surface-treated. For example, the surface treatment may increase a density of upper portions of the second gap-fill insulating layers **101** on which the surface treatment has been performed.

[0079] Referring to FIG. **9**B, second liner patterns **100**A may be formed by etching an upper surface of the second liner layer **100**. The portions of the second liner layer **100** formed on the upper surfaces of the hard mask patterns **99** may be removed. As an example, the second liner layer **100** (e.g., the surface-treated portions of the second linear layer **100**) may be selectively etched using an etching selectivity between the second liner layer **100** and the second gap-fill insulating layer **101**. By soft-etching the second liner layer **100**, the hard mask pattern **99** may be exposed. In addition, a polymer, which is an etching byproduct, may be deposited on surfaces of the second gap-fill insulating layers **101**.

[0080] Referring to FIG. **9**C, an opening OP may be formed by removing the hard mask pattern **99**. For example, each of a plurality of openings OP may be formed between an adjacent pair of the second gap-fill insulating layers **101** in the first direction I by removing a corresponding one of the hard mask patterns **99**. As the surface of the second gap-fill insulating layer **101** is strengthened by the surface treatment, the hard mask pattern **99** may be selectively etched under a condition in which an etching selectivity with respect to the hard mask pattern **99** is high. In a process of

etching the hard mask pattern **99**, the polymer deposited on the surface of the second gap-fill insulating layer **101** may protect the second gap-fill insulating layer **101**.
[0081] Subsequently, a process such as forming a second column line in the opening OP may be performed.
[0082] According to the method described above, the surfaces of the second gap-fill insulating layers **101** may be strengthened by the surface treatment process. In addition, in the process of etching the hard mask patterns **99**, the polymer deposited on the surface of surfaces of the second gap-fill insulating layers **101** may further protect the second gap-fill insulating layers **101**. Accordingly, the hard mask pattern **99** may be selectively etched, and the opening OP exposing the first column line **98** may be formed.
[0083] Although some embodiments according to the technical idea of the present disclosure have been described above with reference to the accompanying drawings, various embodiments of the present disclosure are not limited to the above-described embodiments. Various types of substitutions, modifications, changes, and combinations for the embodiments may be made by those skilled in the art, to which the present disclosure pertains, and these substitutions, modifications, changes, and combinations belong to the scope of the present disclosure.

Claims

1. A semiconductor device comprising: a first row line extending in a first direction; a first column line extending in a second direction that intersects the first direction; a first memory cell connected between the first row line and the first column line; a second column line located above the first column line and extending in the second direction; and a first resistance pattern located between the first column line and the second column line.
2. The semiconductor device of claim 1, further comprising a liner pattern covering a sidewall of the first memory cell, a sidewall of the first column line, and a portion of a sidewall of the second column line.
3. The semiconductor device of claim 2, wherein the second column line comprises: a first portion covered by the liner pattern and having a first width; and a second portion having a greater width than the first portion.
4. The semiconductor device of claim 3, wherein the first resistance pattern covers a bottom surface and a sidewall of the first portion.
5. The semiconductor device of claim 1, further comprising a liner pattern covering a sidewall of the first memory cell, a sidewall of the first column line, and a sidewall of the second column line.
6. The semiconductor device of claim 5, wherein an upper surface of the second column line and an upper surface of the liner pattern are located at substantially the same level.
7. The semiconductor device of claim 1, wherein the first resistance pattern extends along a sidewall of the second column line.
8. The semiconductor device of claim 1, wherein the first resistance pattern includes a material having a higher resistivity than the first column line and the second column line.
9. The semiconductor device of claim 8, wherein the first resistance pattern includes at least one of WN, WSiN, MoN, MoSiN, TiN, or TiCl.sub.4.
10. The semiconductor device of claim 1, wherein the second column line is electrically connected to the first column line through the first resistance pattern.
11. The semiconductor device of claim 1, further comprising a second resistance pattern located between the first memory cell and the first column line.
12. The semiconductor device of claim 1, further comprising: a second row line extending in the first direction; and a second memory cell connected between the second column line and the second row line.
13. The semiconductor device of claim 12, further comprising a third resistance pattern located

between the second column line and the second memory cell.
14. The semiconductor device of claim 1, wherein the first resistance pattern extends along an upper surface of the first column line and a lower surface of the second column line.
15. A method of manufacturing a semiconductor device, the method comprising: forming a cell line that extends in a first direction and includes a variable resistance layer; forming a first conductive layer above the cell line; forming a hard mask pattern on the first conductive layer, the hard mask pattern extending in a second direction that intersects the first direction; forming a first column line by etching the first conductive layer using the hard mask pattern, the first column line extending in the second direction; forming a first memory cell by etching the cell line; forming liner patterns on sidewalls of the first memory cell, sidewalls of the first column line, and sidewalls of the hard mask pattern; forming an opening between the liner patterns by removing the hard mask pattern; forming a resistance pattern in the opening; and forming a second column line in the opening.
16. The method of claim 15, wherein the resistance pattern is formed along a surface of the first column line and surfaces of the liner patterns exposed through the opening.
17. The method of claim 15, wherein the forming of the second column line comprises: forming a second conductive layer to fill the opening; and forming the second column line by etching the second conductive layer, the second column line including a first portion located inside the opening and a second portion located outside the opening, the second portion having a greater width than the first portion.
18. The method of claim 15, wherein the forming of the second column line comprises: forming a second conductive layer to fill the opening; and forming the second column line by planarizing the second conductive layer.
19. The method of claim 15, wherein the resistance pattern includes a material having a higher resistivity than the first column line and the second column line.
20. The method of claim 19, wherein the resistance pattern includes at least one of WN, WSiN, MoN, MoSiN, TiN, or TiCl.sub.4.